CODI-GPT-2 · Steering experiments summary

Question

Are operator/digit/magnitude probe directions causal?
Concretely: at the (pos, layer) cell where each is most decodable, fit a multinomial probe, take its row vector for class c as a steering direction, scale by alpha, add to the residual at the same cell during inference. Measure the effect on the model's emitted answer.

Steering vector convention

v = sigma * (W[c_target] - W[c_source]) in raw residual space, where sigma is the StandardScaler std and W is the LR coefficient matrix.

Crucial caveat about "changed"

Most plots show TOTAL OUTPUTS CHANGED — i.e., the steered output differs from baseline at all. This is NOT the same as 'output now matches the target class'. We also include targeted hit-rate plots to disentangle: did the model move toward the requested digit/op?

What 'changed' means here

changed_full

Number of examples where the steered output's parsed integer differs from the baseline output's parsed integer. Counts perturbation, not intent.

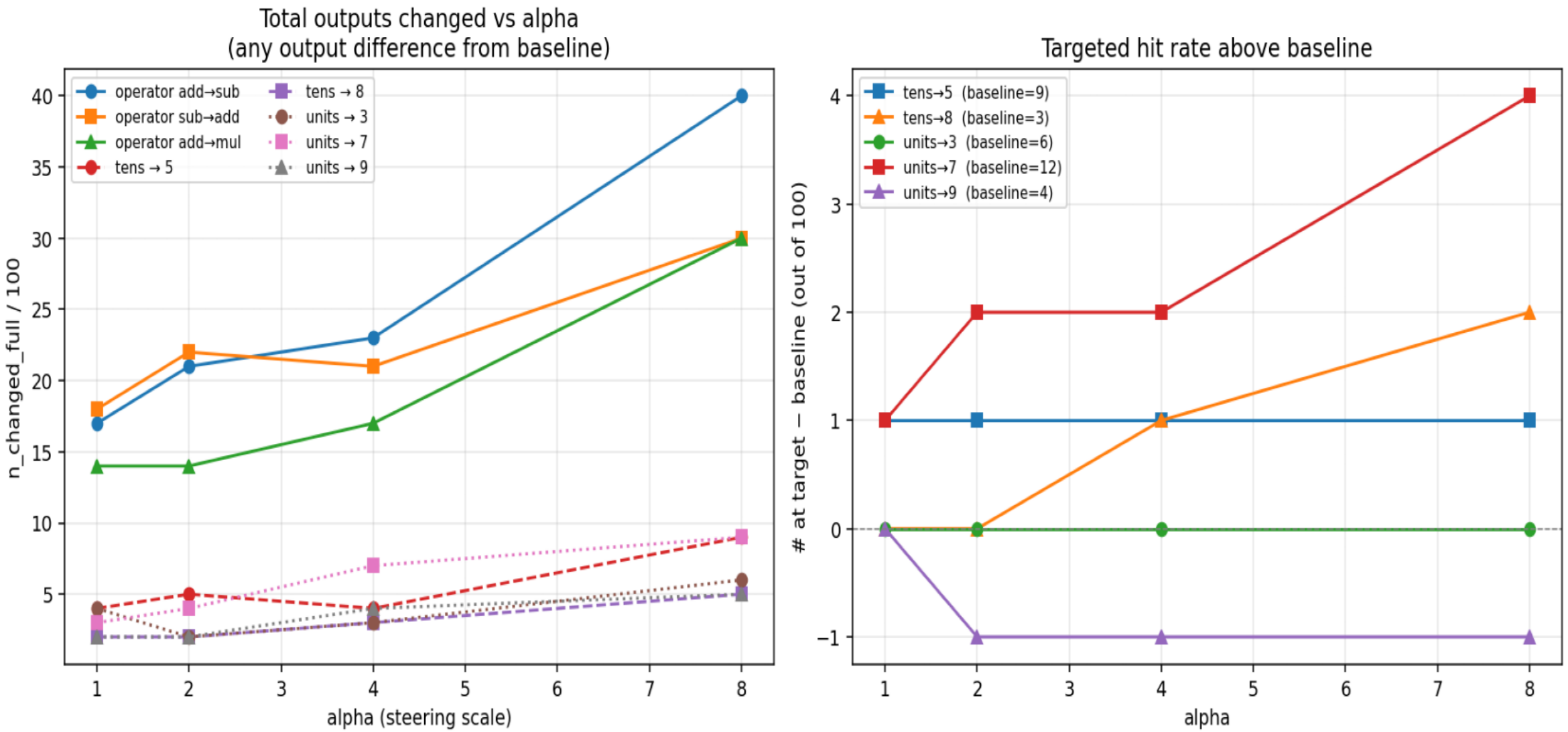
targeted hit-rate (where applicable)

Number whose specific digit (units or tens) equals the requested target, minus the baseline count of that same target. Lift > 0 means causal.

Headline finding when both are looked at

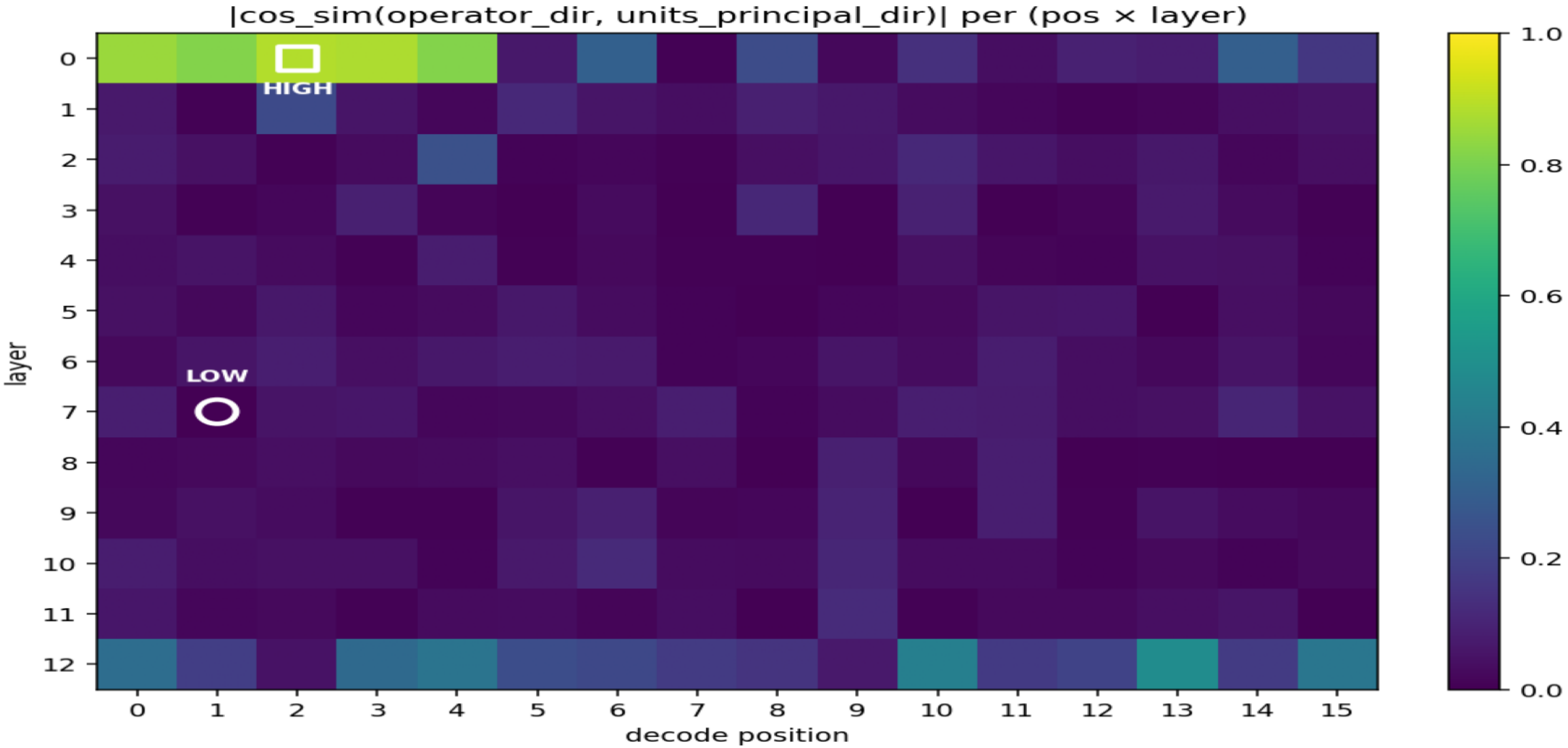
- operator add->sub changed_full=40/100 at alpha=8 but no operator-match is recoverable (model's outputs are nonsense to begin with).
- tens/units sweeps: targeted hit lift +0 to +4 across all alphas. That is roughly noise — the steering moves outputs but not toward the requested digit.
- magnitude: median |int| moves toward the target bucket (-26% / +170%). Direction-correct effect on output magnitude, even if the bucket classification rate barely shifts.

Operator / tens / units steering vs alpha



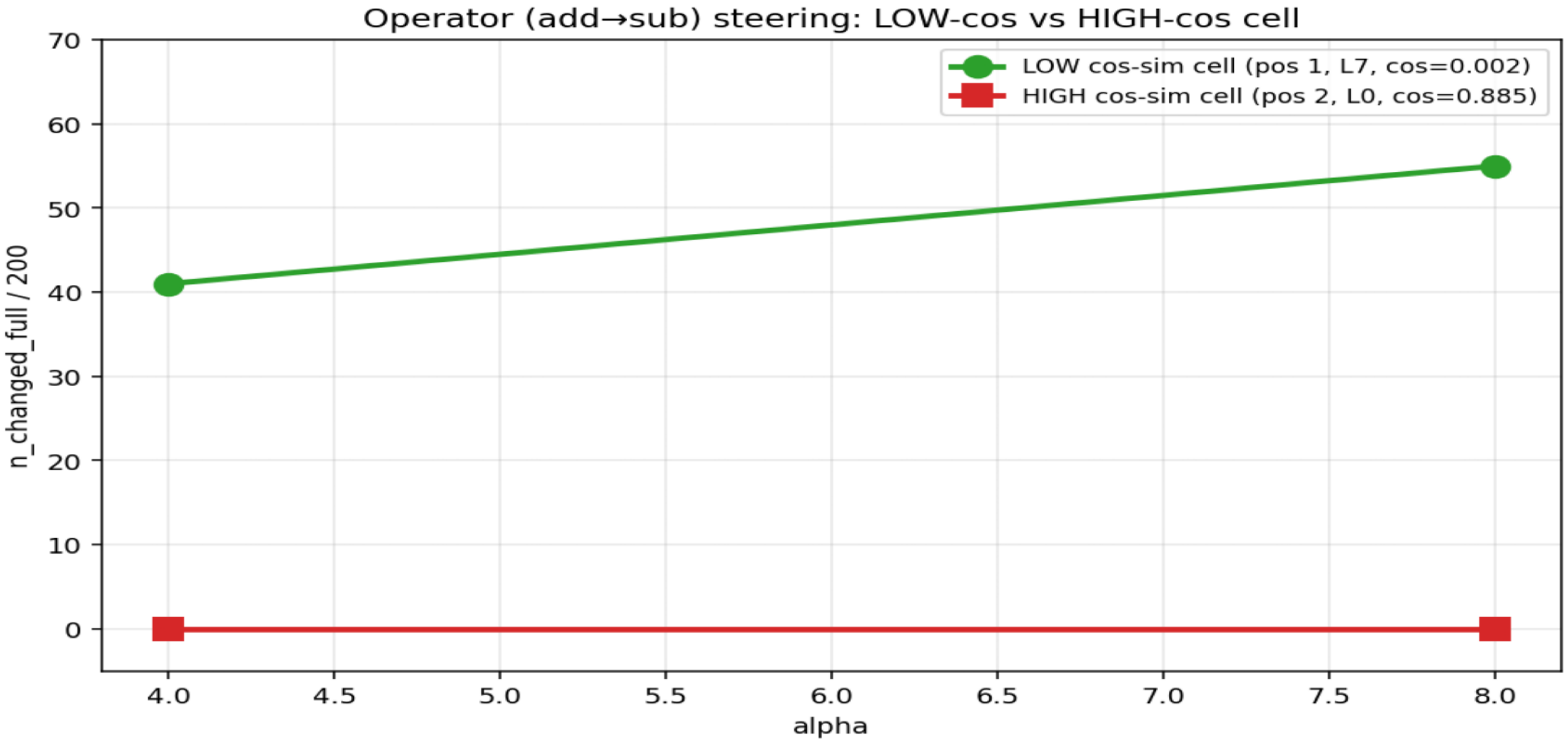
0 at alpha=8. Tens/units stay at <10 across the alpha range — barely steerable. Right: targeted hit rate above baseline. Even strong alphas give +0 to +4 — essentially noise. Operator probes are 91% accurate b

Cos-sim conditioning: where steering works



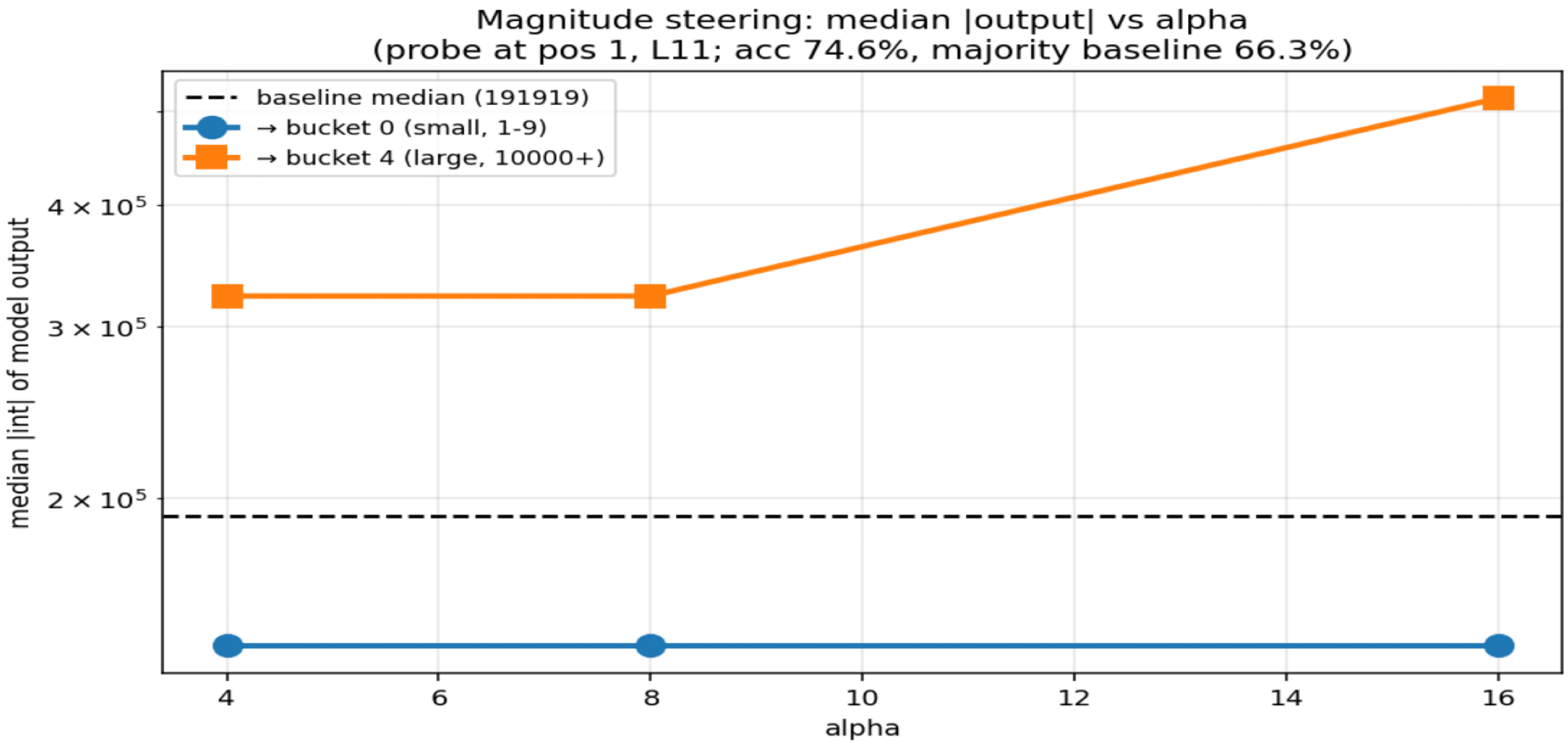
s_sim(operator-direction, units-principal-direction)|. Layer 0 (the embedding) is universally tangled (cos~0.9). Cells at layers 6-12 and positions 1-7 are nearly orthogonal (cos<0.05). LOW cell selected at pos 1 lay

Cos-sim conditioning result: LOW vs HIGH cell



l directions), operator steering changes 41-55 of 200 outputs. At the HIGH-cos cell (tangled directions), the SAME steering vector produces ZERO change. The probe direction is causal only where it's geometrically

Magnitude steering: median |output| vs alpha



et 0) reduces median |int| from 192k to 141k (~26%). Steering toward 'large' (bucket 4) grows it from 192k to 515k (+170%). Direction-correct effect at all alphas tested. Asymmetric because the model's prior is a

What this tells us about computation

Operator

- Probe accuracy 91% — strong linear signal everywhere.
- Causally steerable ONLY at (pos, layer) where its direction is orthogonal to digit directions. Where they're tangled, the same vector has zero behavioral effect.
- Steering perturbs outputs but doesn't recover correct subtraction results — because the model isn't doing arithmetic to begin with.

Specific digits (units, tens)

- Pre-emission probe accuracy ~44%; targeted hit-rate lift = 0 to +4.
- NOT a clean steerable feature in the residual before emission.
- Probe accuracy at later positions (>=7) is 80%+, but that's after the digit has already been emitted into the KV cache — not useful for influencing the answer.

Magnitude (5 buckets by log10)

- Probe accuracy 74.6% at pos 1 L11 (majority baseline 66.3%).
- Steering produces direction-correct shifts in median |output|: -26% toward small, +170% toward large. Real effect at scale level, weak effect at bucket-classification level.

Synthesis

Operator and magnitude exist as steerable linear features (within the right cells). Specific digit values do NOT. The model's 'computation' is best described as: identify operator + commit to a magnitude envelope, then fill in digits one-at-a-time at emission. The digit stream is generated, not pre-computed.